MORPHOLOGY & WORD FORMATION

in Natural Languages

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1. Introduction

What is morphology?

Morphology is the study of the internal structure of words.

It explains how meaningful units combine to create words and express grammatical information.

A morpheme is the smallest linguistic unit that carries meaning or grammatical function.

Example: “unhappiness” = un- + happy + -ness.

2. Morphemes: The Building Blocks

Free and bound morphemes

FREE MORPHEME

Can stand alone as a word.

Examples: book, play, kind

BOUND MORPHEME

Must attach to another form.

Examples: un-, -s, -ed, -ness

ROOT / BASE

The central element carrying core meaning.

Example: “help” in helpful, helpless

Morphemes may express lexical meaning (kind) or grammar (plural -s, past -ed).

3. Types of Morphology

Common ways words change

INFLECTION

Changes grammatical form, not the basic word class.

walk → walked; cat → cats

DERIVATION

Creates a new word or changes its meaning/class.

happy → unhappy; teach → teacher

COMPOUNDING

Combines two or more bases.

black + bird → blackbird

REDUPLICATION

Repeats all or part of a form in some languages.

Examples vary by language and function.

4. Word Formation Processes

How languages create vocabulary

Affixation: add prefixes or suffixes — re + write → rewrite.

Compounding: join words — rain + coat → raincoat.

Conversion: change word class without changing form — email (noun) → to email (verb).

Clipping and blending: advertisement → ad; breakfast + lunch → brunch.

Borrowing: adopt a word from another language.

5. Morphology Across Languages

Languages organize words differently

ANALYTIC

Often uses word order and separate words to express grammar.

Example: I will go.

FUSIONAL

One ending may express several grammatical features.

Example: Spanish verb endings.

In Spanish, the verb hablo means “I speak.” The ending -o indicates the speaker and the present-tense form.

AGGLUTINATIVE

Words may contain several clearly segmentable affixes.

Example: Turkish; many suffixes.

In Turkish, ev means “house,” and evler means “houses.” The suffix -ler indicates plural

These are broad typological patterns; real languages can combine features from more than one type.

6. Why Morphology Matters in NLP

Connecting word forms to meaning

WORD NORMALIZATION

Treat related forms consistently:

connect, connected, connecting

BETTER ANALYSIS

Identify roots, tense, number, and derivation to interpret text.

Examples:
walked → root: walk, tense: past
cats → root: cat, number: plural

RICH-LANGUAGE SUPPORT

Some languages have many word forms and complex prefixes or suffixes. Morphology helps NLP systems process these words.

Morphology is useful in search, translation, sentiment analysis, speech systems, and text classification.

7. Morphology in an NLP Pipeline

A simple example

INPUT

“The cats were playing.”

→

ANALYZE

cat + plural -s

play + -ing

→

REPRESENT

lemma: cat, play

grammar: plural, progressive

→

USE IN TASK

Search, translation, classification, etc.

The system uses morphological clues together with syntax and context; morphology alone does not determine every meaning.

8. Common NLP Techniques

Ways to handle word forms

Stemming: removes word endings using simple rules; it may produce non-words.

Lemmatization: maps a word to its dictionary base form using linguistic information.

Morphological parsing: identifies roots, affixes, and grammatical features.

Subword tokenization: splits words into smaller units for machine-learning models.

Neural language models can learn many patterns, but low-resource and highly inflected languages remain challenging.

9. Challenges and Limitations

Why morphology can be difficult

AMBIGUITY

One form can have several analyses or meanings depending on context.

IRREGULAR FORMS

go → went; mouse → mice do not follow a single simple rule.

DATA SCARCITY

Some languages lack large labeled datasets and reliable analyzers.

Effective systems combine morphological knowledge with context, data, and evaluation.

10. Conclusion

Key takeaways

Morphemes are the smallest meaningful units used to build words.

Inflection, derivation, compounding, and other processes explain how word forms are created.

Morphological analysis helps NLP systems normalize words and interpret grammatical information.

Language diversity, ambiguity, and irregularity make morphology-aware NLP an ongoing research area.

References

Suggested starting points

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